AISC 5020 Mathematics & Statistics for AI

Lecture 2 | Linear Algebra I

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(Recall) AI pipeline

An AI pipeline is typically as follows:



Repeat on training data; evaluate on held-out data.

Question	Mathematical tools
How do we represent measurements?	Vectors and matrices
How does an output change?	Calculus
What is uncertain?	Probability and statistics
How do we improve a model?	Optimization

(Recall) Three broad tasks in AI for science

Task	Desired output	Example
Regression	Number	Predicted concentration
Classification	Category or class probability	Predicted material type
Generative modelling	New structured sample	Candidate molecular structure

The **scientific question** determines what task is to be performed.

Example: For a collection of chemical samples:

- Predict reaction yield: regression
- Predict whether yield exceeds a chosen threshold: classification
- Propose a new candidate sample: generative modelling

The same raw data may be used for different tasks.

(Recall) Review of some basic mathematical concepts

- Basic arithmetic and functions
- Set theory
- Probability
- Vectors and matrices
- Calculus

Topics in this lecture

We will introduce more concepts in linear algebra relevant to AI:

- Scalars, vectors, matrices and tensors
- Dot products, cosine similarity
- L1/L2 norms, distances, orthogonality
- Applications: similarity, embeddings

Why do we need linear algebra in AI?

Linear algebra is the mathematical backbone of AI, providing the structures and rules needed to represent, transform, and process massive amounts of data efficiently

- **Data Representation**: Turns real-world data into a suitable mathematical representation
- **Data Transformations**: Many AI models (regression, neural networks, etc.) utilize linear algebra techniques to build transformations/solvers
- **Data Similarity**: Linear algebra provides methods for evaluating similarity between data points
- **Data Reduction**: Linear algebra provides efficient methods for simplifying giant datasets while keeping the most important information

Scalars, vectors, matrices and tensors

Scalar: A numeric value

Examples: $a = 1$, $b = 2.34$, $c = -5678$, temperature, model weight, loss value. ...

Vector: an ordered list of numbers of the form $(a_1, a_2, \cdots, a_n)$ or $\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$

Examples: $\begin{pmatrix} 1 & 3 \end{pmatrix}$, $\begin{pmatrix} 3 \\ -1 \\ 4 \\ -1.59 \end{pmatrix}$

Scalars, vectors, matrices and tensors

Matrix: A rectangular array of the form

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix},$$

where each entry a_{ij} $(1 \leq i \leq m, 1 \leq j \leq n)$ is a real number.
Example: Below is a 2×3 matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \in \mathbb{R}^{2\times 3}.$$

A_{ij} means row i, column j; e.g., $A_{23} = 6$.

Scalars, vectors, matrices and tensors

Tensor: a multi-dimensional array for more complex data

- 0D Tensor (Scalar): A single number with magnitude but no direction (e.g., 42).
- 1D Tensor (Vector): An array of numbers arranged in a single line (e.g., $[1.2, 3.5, 4.6]$).
- 2D Tensor (Matrix): A traditional grid or table with rows and columns (e.g., $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$).
- 3D Tensor: A cube of numbers (e.g., a color image with dimensions for height, width, and color channels).
- Higher-order (4D, 5D, ...) Tensors: Higher-order structures used for complex datasets (e.g., a video tensor tracking frames, height, width, and color channels).

Example: A 256×256 color image (with red, green, blue channels) might be stored as

$$\mathcal{X} \in \mathbb{R}^{256 \times 256 \times 3}.$$

Example

For each of the following data, give an appropriate representation and the corresponding shape

- 40 samples, each with 12 features
- One 12-feature sample as a column
- 5 time series, each with 100 time points and 4 channels

Example

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- 40 samples, each with 12 features
- One 12-feature sample as a column
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Answer:

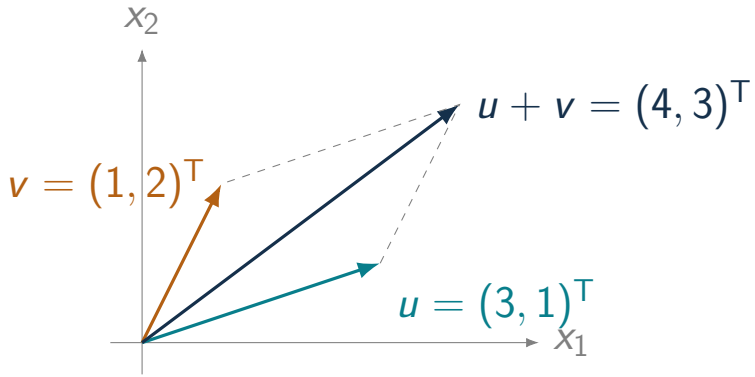
- A 40×12 matrix
- A 12×1 vector
- A $5 \times 100 \times 4$ tensor

Addition and scalar multiplication of vectors and matrices

For vector addition, the vectors must have matching coordinate meanings and lengths. The addition is then coordinate by coordinate, e.g.,

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}.$$

We can also have a geometric interpretation:



The parallelogram and tip-to-tail constructions give the same vector sum. The geometry agrees with the coordinate calculation.

Addition and scalar multiplication of vectors and matrices

Scalar multiplication of a vector serves as a rescaling, e.g.,

$$2\begin{bmatrix}3\\1\end{bmatrix} = \begin{bmatrix}6\\2\end{bmatrix}, \qquad -\begin{bmatrix}3\\1\end{bmatrix} = \begin{bmatrix}-3\\-1\end{bmatrix}.$$

Note:

- A positive scalar preserves direction and changes length.
- A negative scalar also reverses direction.
- Multiplication by zero produces the zero vector, which has no direction.

Addition and scalar multiplication of vectors and matrices

Similarly, we can do addition and scalar multiplication of matrices provided that they have matching shapes, e.g.,

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 0 & -1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 5 & 5 \end{bmatrix}, \quad 2 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}.$$

The same rule applies to tensors.
In practice, it is common to **reshape your data representation** into vectors to simplify subsequent computations and operations.
Example:

- Reshaping a 2×3 matrix as a 6×1 vector
- Reshaping a $m \times n \times p$ tensor as a $(mnp) \times 1$ vector

Therefore, we will focus on **vectors** in many of our subsequent theories and computations.

Linear combination of vectors

A **linear combination** of two vectors $\mathbf{u}, \mathbf{v}$ is in the form of

$$a\mathbf{u} + b\mathbf{v}$$

where a, b are scalars.

Example: For $\mathbf{u} = (1, 0)^T$, $\mathbf{v} = (0, 1)^T$, a linear combination is

$$3\mathbf{u} - 2\mathbf{v} = (3, -2)^T.$$

Example: If $a_i \geq 0$ and $\sum_i a_i = 1$, then $\sum_i a_i \mathbf{x_i}$ is said to be a weighted average. Note that weighted averages of vectors are also linear combinations. For $\mathbf{u} = (2, 4)^T$ and $\mathbf{v} = (6, 0)^T$,

$$\frac{3}{4}\mathbf{u} + \frac{1}{4}\mathbf{v} = (3, 3)^T.$$

Example

Let $\mathbf{u} = (2, 0, 1)^{\mathsf{T}}$ and $\mathbf{v} = (0, 4, 1)^{\mathsf{T}}$. Compute $\mathbf{u} + \mathbf{v}$, $2\mathbf{u} - \mathbf{v}$ and $\frac{1}{2}(\mathbf{u} + \mathbf{v})$.

Example

Let $\mathbf{u} = (2, 0, 1)^T$ and $\mathbf{v} = (0, 4, 1)^T$. Compute $\mathbf{u} + \mathbf{v}$, $2\mathbf{u} - \mathbf{v}$ and $\frac{1}{2}(\mathbf{u} + \mathbf{v})$.

Answer:

$$\mathbf{u} + \mathbf{v} = (2, 0, 1)^T + (0, 4, 1)^T = (2 + 0, 0 + 4, 1 + 1)^T = (2, 4, 2)^T$$

$$2\mathbf{u} - \mathbf{v} = 2(2, 0, 1)^T - (0, 4, 1)^T = (4, -4, 1)^T$$

$$\frac{1}{2}(\mathbf{u} + \mathbf{v}) = \frac{1}{2}(2, 0, 1)^T + \frac{1}{2}(0, 4, 1)^T = (1, 2, 1)^T.$$

Linear combinations and matrix-vector multiplication

Note that linear combinations of vectors can also be expressed as a matrix-vector multiplication. More specifically, we can put all vectors (in the form of column vectors) into a matrix A. Then the matrix-vector multiplication Ax gives a linear combination of the vectors.

Example: Let $\mathbf{u} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$. We can form a matrix

$$A = [\mathbf{u}\ \mathbf{v}] = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}.$$

Then, for the linear combination $2\mathbf{u} + \mathbf{v}$, we can set $x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and write

$$A\mathbf{x} = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}.$$

Linear map

A map T is said to be linear if for any compatible vectors $\mathbf{x}, \mathbf{y}$ and any scalars α, β, we have

$$T(\alpha\mathbf{x} + \beta\mathbf{y}) = \alpha T(\mathbf{x}) + \beta T(\mathbf{y})$$

Example: The matrix maps $T(\mathbf{x}) = A\mathbf{x}$ are linear maps, as we have

$$A(\alpha\mathbf{x} + \beta\mathbf{y}) = A(\alpha\mathbf{x}) + A(\beta\mathbf{y}) = \alpha(A\mathbf{x}) + \beta(A\mathbf{y}).$$

This is why matrix algebra naturally describes additive signal mixing and many discretized scientific systems.

More generally, rotations, reflections, and scaling transformations are also examples of linear maps.

Linear map

We remark that the following affine map is NOT a linear map

$$f(\mathbf{x}) = A\mathbf{x} + \mathbf{b}, \text{ with } \mathbf{b} \neq \mathbf{0}.$$

Reason: By definition of linear map, we should have $T(\alpha\mathbf{u} + \beta\mathbf{v}) = \alpha T(\mathbf{u}) + \beta T(\mathbf{v})$ for any $\mathbf{u}, \mathbf{v}, \alpha, \beta$.
Putting $\mathbf{u} = \mathbf{v} = \mathbf{0}$ and $\alpha = \beta = 1$, we have

$$T(\mathbf{0}) = T(\mathbf{0}) + T(\mathbf{0}) \Rightarrow T(\mathbf{0}) = \mathbf{0}.$$

However, for the affine map above, we have

$$f(\mathbf{0}) = A\mathbf{0} + \mathbf{b} = \mathbf{b} \neq \mathbf{0}$$

Therefore, the map is not linear.

Linearly independent vectors

A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ are linearly independent if and only if none of them can be written as a linear combination of the other vectors. Equivalently, $\{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ are linearly independent if and only if

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n = 0 \text{ implies } a_1 = a_2 = \cdots = a_n = 0.$$

Examples:

- $\begin{pmatrix}2\\3\end{pmatrix}, \begin{pmatrix}1\\1\end{pmatrix}$ are linearly independent
- $\begin{pmatrix}2\\3\end{pmatrix}, \begin{pmatrix}1\\1\end{pmatrix}, \begin{pmatrix}3\\4\end{pmatrix}$ are linearly dependent as $\begin{pmatrix}3\\4\end{pmatrix} = \begin{pmatrix}2\\3\end{pmatrix} + \begin{pmatrix}1\\1\end{pmatrix}$

Property: A square matrix is invertible if and only if all its columns (or rows) are linearly independent vectors.

Solving linear systems

Recall that one fundamental task in linear algebra is to solve systems of linear equations:

$$A\mathbf{x} = \mathbf{b}$$

A: $m \times n$ matrix, $\mathbf{x}$: $n \times 1$ vector, $\mathbf{b}$: $m \times 1$ vector

- $m = n$: **Square** linear system
  - Same number of equations (rows) and unknowns (columns)
  - Unique solution if all rows/columns are linearly independent
- $m > n$: **Overdetermined** system
  - More equations than unknowns
  - No true solution, can only solve in a least-squares sense
- $m < n$: **Underdetermined** system
  - More unknowns than equations
  - Multiple solutions, can choose the "best" solution based on some extra criterion

Gaussian elimination and LU factorization

A common approach to solve $A\mathbf{x} = \mathbf{b}$ is the **Gaussian elimination**, which involves using three types of elementary row operations to modify the matrix until the lower left-hand corner of the matrix is filled with zeros as much as possible:

- multiplying a row by a nonzero number,
- adding a multiple of one row to another row, and
- swapping two rows.

Conceptually, the process is as follows:

$$A = \begin{bmatrix} \times & \times & \dots & \times & \times \\ \times & \times & \dots & \times & \times \\ \times & \times & \dots & \times & \times \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \times & \times & \dots & \times & \times \end{bmatrix} \xrightarrow{L_1} \begin{bmatrix} \times & \times & \dots & \times & \times \\ 0 & \times & \dots & \times & \times \\ 0 & \times & \dots & \times & \times \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & \times & \dots & \times & \times \end{bmatrix} \xrightarrow{L_2} \begin{bmatrix} \times & \times & \dots & \times & \times \\ 0 & \times & \dots & \times & \times \\ 0 & 0 & \dots & \times & \times \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & \times & \times \end{bmatrix} \xrightarrow{L_3} \dots \xrightarrow{L_{n-1}} \begin{bmatrix} \times & \times & \dots & \times & \times \\ 0 & \times & \dots & \times & \times \\ 0 & 0 & \dots & \times & \times \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 0 & \times \end{bmatrix} = U$$

Gaussian elimination and LU factorization

The three types of operations can be represented using multiplication by elementary matrices:

(1) (multiplying row i by m) $L = \begin{bmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & & & & \\ & & & m & & & \\ & & & & 1 & & \\ & & & & & \ddots & \\ & & & & & & 1 \end{bmatrix}$, where m is the (i,i)-th entry

(2) (adding row j multiplied by m to row i) $L = \begin{bmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & & & & \\ & & & \ddots & & & \\ & & m & & 1 & & \\ & & & & & \ddots & \\ & & & & & & 1 \end{bmatrix}$, where m is the (i,j)-th entry

(3) (swapping two rows i, j) $P_{i \leftrightarrow j} = \begin{bmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 0 & & 1 & & \\ & & & \ddots & & & \\ & & 1 & & 0 & & \\ & & & & & \ddots & \\ & & & & & & 1 \end{bmatrix}$, where $\begin{cases} \text{the } (i,i)\text{-th entry} = 0 \\ \text{the } (j,j)\text{-th entry} = 0 \\ \text{the } (i,j)\text{-th entry} = 1 \\ \text{the } (j,i)\text{-th entry} = 1 \end{cases}$

Gaussian elimination and LU factorization

If we only use Type (1) and Type (2) operations above (i.e., without row swapping), we can summarize the process as a series of matrix operations:

$$A \to L_1 A \to L_2 L_1 A \to \cdots \to L_{n-1} \cdots L_2 L_1 A = U$$

This gives the **LU factorization** $\boxed{A = LU}$ where

- $L = (L_1 L_2 \cdots L_{n-1})^{-1}$ is an $n \times n$ lower triangular matrix with
  - all diagonal entries $= 1$
  - all subdiagonal entries $= -$(corresponding entries in $L_1, \ldots, L_{n-1}$)
- U is an $n \times n$ upper triangular matrix

Gaussian elimination and LU factorization

If $A = LU$, then

$$A\mathbf{x} = \mathbf{b} \iff LU\mathbf{x} = \mathbf{b}$$

can be easily solved in two steps:

1. Solving $L\mathbf{y} = \mathbf{b}$ for $\mathbf{y}$ using forward substitution
   - Solve the equations from top to bottom
   - The first equation contains only one unknown variable y_1
   - The second equation contains y_1, y_2, so we can solve it after getting y_1 above
   - Similarly, we can solve for $y_3, y_4, \ldots, y_n$ one by one
2. Solving $U\mathbf{x} = \mathbf{y}$ for $\mathbf{x}$ using backward substitution
   - Solve the equations from bottom to top
   - The last equation contains only one unknown variable x_n
   - The second equation contains x_{n-1}, x_n, so we can solve it after getting x_{n-1} above
   - Similarly, we can solve for $x_{n-2}, x_{n-3}, \ldots, x_1$ one by one

Examples

To solve $\begin{pmatrix} 2 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix}$ using forward substitution, we have

$$x_1 = 4/2 = 2$$

$$x_2 = 2 - 2x_1 = -2$$

$$x_3 = 3 - 3x_1 - 2x_2 = 1$$

To solve $\begin{pmatrix} 3 & 2 & 5 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix}$ using backward substitution, we have

$$x_3 = 9/3 = 3$$

$$x_2 = 8 - 2x_3 = 2$$

$$x_1 = (7 - 2x_2 - x_3)/3 = -4$$

Gaussian elimination and LU factorization

Remark:

- LU factorization is one of the most standard methods for solving $A\mathbf{x} = \mathbf{b}$, but the factorization $A = LU$ is not always possible!
- More generally, if we also utilize the third type of row operations mentioned above (row swapping, represented as permutation matrix $P_{i\leftrightarrow j}$), we can always get a PLU factorization
- Process:

$$A \to P_1 A \to L_1 P_1 A \to \cdots \to L_{n-1} P_{n-1} \cdots L_2 P_2 L_1 P_1 A = U$$

- We have $PA = LU$ (or equivalently, $A = PLU$) where
  - L is an $n \times n$ lower triangular matrix with all diagonal entries $= 1$ (Note: L is not equal to $(L_1 L_2 \cdots L_{n-1})^{-1}$ in this case)
  - U is an $n \times n$ upper triangular matrix
  - P is an $n \times n$ permutation matrix (Note: $P = P_{n-1} P_{n-2} \cdots P_1$)

Gaussian elimination and LU factorization

Note that the PLU factorization can also help us solve $A\mathbf{x} = \mathbf{b}$

- If $PA = LU$, then

$$A\mathbf{x} = \mathbf{b} \iff LU\mathbf{x} = P\mathbf{b}$$

- It can then be solved in two steps:
  - Solving $L\mathbf{y} = P\mathbf{b}$ for $\mathbf{y}$ using forward substitution
  - Solving $U\mathbf{x} = \mathbf{y}$ for $\mathbf{x}$ using backward substitution

Remark: Note that many other matrix factorization methods and numerical methods can be considered. For example, we will introduce the QR factorization later this lecture.

Dot product

For any two d-dimensional vectors $\mathbf{u} = (u_1, u_2, \ldots, u_d), \mathbf{v} = (v_1, v_2, \ldots, v_d) \in \mathbb{R}^d$, we define the **dot product** of them as

$$\boxed{\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_d v_d = \sum_{j=1}^{d} u_j v_j}$$

In other words, we multiply the entries in a coordinate-wise manner and add up the products, and the result is a scalar.

We can see that large positive contributions arise where both entries have the same sign and substantial magnitude.

Also,

- If $\mathbf{u}, \mathbf{v}$ are represented as column vector, we can write the dot product as $\boxed{\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^\top \mathbf{v}}$.
- Another notation for dot product is $\boxed{\mathbf{u} \cdot \mathbf{v} = \langle \mathbf{u}, \mathbf{v} \rangle}$

All the above notations are commonly used in linear algebra.

Example

We compute the dot product for the following vectors in three dimensions:

$$\mathbf{u} = (1, 2, -1)^{\mathsf{T}}, \qquad \mathbf{v} = (3, 0, 2)^{\mathsf{T}}.$$

We have

$$\mathbf{u} \cdot \mathbf{v} = 1(3) + 2(0) + (-1)(2) = 1.$$

Alternatively, we have

$$\mathbf{u}^{\mathsf{T}}\mathbf{v} = \begin{pmatrix} 1 & 2 & -1 \end{pmatrix} \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix} = 1(3) + 2(0) + (-1)(2) = 1.$$

From the above, we can see that:

- Some negative products may cancel part of the positive contributions.
- The overall dot product is a signed comparison of two vectors, not just a count of matching nonzero entries.

Dot products as weighted sums

Another common use of dot products (in regression, neural networks, ...) is computing a weighted sum of some numbers

Example: With $\mathbf{x} = (2, 1, 3)^\mathsf{T}$ and $\mathbf{w} = (0.5, -1, 2)^\mathsf{T}$, we have

$$\mathbf{w} \cdot \mathbf{x} = 0.5(2) - 1(1) + 2(3) = 6.$$

The weights $\mathbf{w}$ can amplify, ignore or reverse feature contributions.

More properties of dot product

From the definition of the dot product $\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_d v_d$, we have the following properties:

- (Symmetry) The dot product of $\mathbf{u}$ and $\mathbf{v}$ is equal to the dot product of $\mathbf{v}$ and $\mathbf{u}$:

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$$

  (i.e., the order does not matter)
- (Linearity) For any vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and any scalar a, b, we have

$$(a\mathbf{u} + b\mathbf{v}) \cdot \mathbf{w} = a(\mathbf{u} \cdot \mathbf{w}) + b(\mathbf{v} \cdot \mathbf{w})$$

  (i.e., we can decompose the vector and handle the computation part by part)
- (Positive-definiteness) For any vector $\mathbf{x}$, we have

$$\mathbf{x} \cdot \mathbf{x} \geq 0$$

  and the value is exactly 0 if any only if $\mathbf{x} = (0, 0, \ldots, 0)$ (i.e, a zero vector)

Geometric interpretation of dot product

For any two nonzero d-dimensional vectors $\mathbf{u}, \mathbf{v}$, we can express the dot product using the angle θ between the two vectors

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \, \|\mathbf{v}\| \cos\theta, \qquad 0 \leq \theta \leq \pi.$$

Here, $\|\mathbf{u}\| = \sqrt{u_1^2 + \cdots + u_d^2}$ is the Euclidean norm.

Interpretations:

- Positive dot product: angle less than 90°.
- Zero: perpendicular directions.
- Negative: angle greater than 90°.

A large dot product can reflect large magnitudes as well as aligned directions.

Cosine similarity

Given two n-dimensional vectors $\mathbf{u}, \mathbf{v}$, we can define the **cosine similarity** of them using the cosine of the angle between the two vectors:

$$\text{cosine similarity} = \cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2} = \frac{u_1v_1 + \cdots + u_nv_n}{\sqrt{u_1^2 + \cdots + u_n^2}\cdot\sqrt{v_1^2 + \cdots + v_n^2}}$$

Properties:

- It lies in $[-1, 1]$ for nonzero real vectors.
- -1 meaning exactly opposite
- $+1$ meaning exactly proportional
- 0 indicating orthogonality (no correlation)
- In-between values indicate intermediate similarity or dissimilarity.

Cosine similarity

$$\text{cosine similarity} = \cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|_2\,\|\mathbf{v}\|_2} = \frac{u_1v_1 + \cdots + u_nv_n}{\sqrt{u_1^2 + \cdots + u_n^2}\cdot\sqrt{v_1^2 + \cdots + v_n^2}}$$

Key features:

- **Ignores Magnitude**: Focuses purely on direction and angle, disregards the scale or length of the vectors
- **High-Dimensional Utility**: Works very well in large vector spaces

Common uses:

- **Text Analysis**: Comparing word frequency or semantic embeddings between documents.
- **Recommendation Systems**: Matching user preferences based on ratings.
- **Information Retrieval**: Finding relevant search results for a query

Example

Let $\mathbf{u} = (1, 3)^\mathsf{T}$ and $\mathbf{v} = (2, 6)^\mathsf{T}$. We have

$$\mathbf{u} \cdot \mathbf{v} = 1 \cdot 2 + 3 \cdot 6 = 20,$$
$$\|\mathbf{u}\|_2 = \sqrt{1^2 + 3^2} = \sqrt{10},$$
$$\|\mathbf{v}\|_2 = \sqrt{2^2 + 6^2} = \sqrt{40} = 2\sqrt{10},$$

and so the cosine similarity is

$$\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2} = \frac{20}{\sqrt{10} \times 2\sqrt{10}} = 1$$

The two vectors are exactly proportional.

Example

Let $\mathbf{u} = (1, 1, 0)^{\mathsf{T}}$, $\mathbf{v} = (1, 0, 1)^{\mathsf{T}}$, $\mathbf{w} = (-1, -1, 0)^{\mathsf{T}}$.

- Compute the cosine similarity of $\mathbf{u}$ and $\mathbf{v}$.
- Compute the cosine similarity of $\mathbf{u}$ and $\mathbf{w}$.
- Which pair has opposite directions?

Example

Let $\mathbf{u} = (1, 1, 0)^T$, $\mathbf{v} = (1, 0, 1)^T$, $\mathbf{w} = (-1, -1, 0)^T$.

- Compute the cosine similarity of $\mathbf{u}$ and $\mathbf{v}$.
- Compute the cosine similarity of $\mathbf{u}$ and $\mathbf{w}$.
- Which pair has opposite directions?

Answer:

$$\mathbf{u} \cdot \mathbf{v} = 1 \cdot 1 + 1 \cdot 0 + 0 \cdot 1 = 1, \quad \|\mathbf{u}\|_2 = \|\mathbf{v}\|_2 = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2},$$

and so the cosine similarity of $\mathbf{u}$ and $\mathbf{v}$ is $\dfrac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2} = \dfrac{1}{\sqrt{2} \times \sqrt{2}} = \dfrac{1}{2}$.

Also, we have

$$\mathbf{u} \cdot \mathbf{w} = 1 \cdot (-1) + 1 \cdot (-1) + 0 \cdot 0 = -2, \quad \|\mathbf{w}\|_2 = \sqrt{(-1)^2 + (-1)^2 + 0} = \sqrt{2},$$

and so the cosine similarity of $\mathbf{u}$ and $\mathbf{w}$ is $\dfrac{\mathbf{u} \cdot \mathbf{w}}{\|\mathbf{u}\|_2 \|\mathbf{w}\|_2} = \dfrac{-2}{\sqrt{2} \times \sqrt{2}} = \dfrac{-2}{2} = -1$.

Thus $\mathbf{u}$ and $\mathbf{w}$ point in opposite directions; $\mathbf{u}$ and $\mathbf{v}$ form a 60° angle.

Cauchy–Schwarz inequality

A well-known inequality related to dot product is the **Cauchy–Schwarz inequality**, which gives a bound for dot products.
More specifically, for any $\mathbf{u} = (u_1, \ldots, u_d), \mathbf{v} = (v_1, \ldots, v_d)$, we have

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \, \|\mathbf{u}\| \, .$$

In other words,

$$|u_1 v_1 + \cdots + u_d v_d| \leq \left(\sqrt{u_1^2 + \cdots + u_d^2}\right) \left(\sqrt{v_1^2 + \cdots + v_d^2}\right) .$$

Note:

- This inequality ensures cosine similarity cannot exceed 1 in magnitude.
- Equality holds when the vectors are linearly dependent.

Cauchy–Schwarz inequality

To prove this inequality, we first note that for any scalar t, we have

$$0 \leq \|\mathbf{u} - t\mathbf{v}\|^2.$$

We can further express the above as

$$0 \leq (u_1 - tv_1)^2 + \cdots + (u_d - tv_d)^2 = (u_1^2 - 2tu_1v_1 + t^2v_1^2) + \cdots + (u_d^2 - 2tu_dv_d + t^2v_d^2)$$

By rearranging and regrouping the terms, we have

$$0 \leq (u_1^2 + \cdots + u_d^2) - 2t(u_1v_1 + \cdots + u_dv_d) + t^2(v_1^2 + \cdots + v_d^2) = \|\mathbf{u}\|^2 - 2t(\mathbf{u} \cdot \mathbf{v}) + t^2\|\mathbf{v}\|^2.$$

If $\mathbf{v}$ is a zero vector, both sides of the Cauchy–Schwarz inequality are simply 0 and so the inequality holds. If $\mathbf{v}$ is a non-zero vector, we substitute $t = (\mathbf{u} \cdot \mathbf{v})/\|\mathbf{v}\|^2$ into the above and get

$$0 \leq= \|\mathbf{u}\|^2 - 2\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2}(\mathbf{u} \cdot \mathbf{v}) + \frac{(\mathbf{u} \cdot \mathbf{v})^2}{(\|\mathbf{v}\|^2)^2}\|\mathbf{v}\|^2 = \|\mathbf{u}\|^2 - \frac{(\mathbf{u} \cdot \mathbf{v})^2}{\|\mathbf{v}\|^2}$$

Rearranging gives $(\mathbf{u} \cdot \mathbf{v})^2 \leq \|\mathbf{u}\|^2\|\mathbf{v}\|^2$. Taking square roots proves the inequality.

Vector norm

Recall that: for a vector $\mathbf{x} = (x_1, x_2, \cdots, x_n)^T$, its ***p*-norm** is defined as

$$\|\mathbf{x}\|_p = \left(|x_1|^p + |x_2|^p + \cdots + |x_n|^p\right)^{\frac{1}{p}}, \quad 1 \leq p \leq \infty.$$

In particular, all p-norms satisfy the three requirements of a norm:

1. $\|\mathbf{x}\|_p \geq 0$, and $\|\mathbf{x}\|_p = 0$ if and only if $\mathbf{x} = \mathbf{0}$, where $\mathbf{0}$ is the zero vector.
2. $\|\alpha\mathbf{x}\|_p = |\alpha| \cdot \|\mathbf{x}\|_p$, for all $\alpha \in \mathbb{R}$.
3. (Triangle Inequality): $\|\mathbf{x} + \mathbf{y}\|_p \leq \|\mathbf{x}\|_p + \|\mathbf{y}\|_p$.

These properties formalize a consistent notion of length. Different norms emphasize different aspects of a vector.

The L2 norm

The L2 norm is also known as the Euclidean norm, defined as follows: For any n-dimensional vector $\mathbf{x} = (x_1, x_2, \ldots, x_n)$, we have

$$\|\mathbf{x}\|_2 = \sqrt{x_1^2 + \cdots + x_n^2}.$$

We can also express it using the dot product:

$$\|\mathbf{x}\|_2 = \sqrt{\mathbf{x} \cdot \mathbf{x}}.$$

Examples:

- For $\mathbf{x} = (3, 4)^\mathsf{T}$, $\|\mathbf{x}\|_2 = \sqrt{3^2 + 4^2} = 5$.
- For $\mathbf{x} = (1, -2, 2)^\mathsf{T}$, $\|\mathbf{x}\|_2 = \sqrt{1^2 + (-2)^2 + 2^2} = 3$.

Features:

- The square removes sign of each component, and the square root restores the original units.
- The value will be dominated by the larger components because of the use of square.

The L1 norm

The L1 norm is another commonly used norm. For any n-dimensional vector $\mathbf{x} = (x_1, x_2, \ldots, x_n)$, we have

$$\|\mathbf{x}\|_1 = |x_1| + \cdots + |x_n|.$$

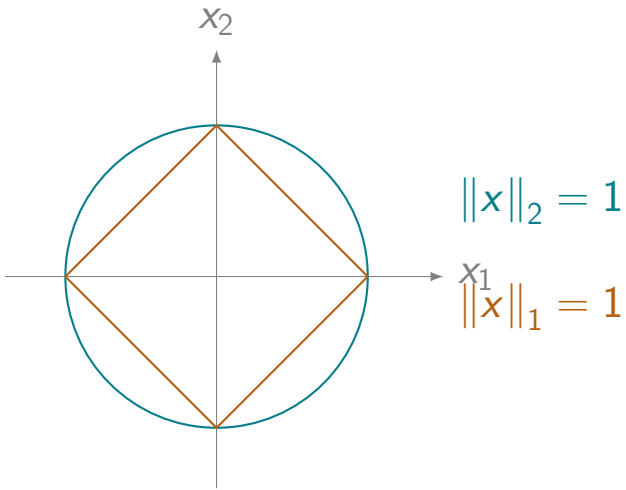
Examples:

- For $\mathbf{x} = (3, 4)^\mathsf{T}$, $\|\mathbf{x}\|_1 = |3| + |4| = 7$.
- For $\mathbf{x} = (1, -2, 2)^\mathsf{T}$, $\|\mathbf{x}\|_1 = |1| + |-2| + |2| = 5$.

Features:

- The absolute function removes sign of each component.
- Each coordinate contributes linearly in its absolute magnitude.

Geometric interpretation of L1 and L2 norms



In two dimensions, the L2 unit ball is a circle, and the L1 unit ball is a rotated square. Both pass through $(1, 0)$ and $(0, 1)$ but treat diagonal directions differently.

Distances

Using the definition of norms, we can easily evaluate the distance between points.
Specifically, the distance between two points $\mathbf{x}, \mathbf{y}$ under any chosen norm can be given by

$$\boxed{d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|}.$$

In particular, if we use the L2 norm, we have

$$d_2(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_2 = \sqrt{(x_1 - y_1)^2 + \cdots + (x_d - y_d)^2}.$$

If we use the L1 norm, we have

$$d_1(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_1 = |x_1 - y_1| + \cdots + |x_d - y_d|.$$

Example: For $\mathbf{x} = (1, 2)^\mathsf{T}$ and $\mathbf{y} = (4, 6)^\mathsf{T}$,

$$d_2(\mathbf{x}, \mathbf{y}) = \sqrt{(1 - 4)^2 + (2 - 6)^2} = 5, \qquad d_1(\mathbf{x}, \mathbf{y}) = |1 - 4| + |2 - 6| = 7.$$

Triangle inequality

Recall that the vector norms satisfy the triangle equality $\|\mathbf{u}+\mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ by definition. Consequently, for any three points $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^d$, we have

$$d(\mathbf{x}, \mathbf{z}) = \|\mathbf{x}-\mathbf{z}\| = \|(\mathbf{x}-\mathbf{y})+(\mathbf{y}-\mathbf{z})\| \leq \|\mathbf{x}-\mathbf{y}\| + \|\mathbf{y}-\mathbf{z}\| = d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z})$$

In other words, the triangle inequality for distances can be expressed as:

$$\boxed{d(\mathbf{x}, \mathbf{z}) \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z})}$$

- A direct displacement cannot be longer than travelling through an intermediate point under a norm-based distance.
- This property helps organize nearest-neighbor search and geometric reasoning for many AI problems.

Example

Let $\mathbf{x} = (2, -1, 3)^{\mathsf{T}}$ and $\mathbf{y} = (-1, 1, 3)^{\mathsf{T}}$. Compute $d_1(\mathbf{x}, \mathbf{y})$ and $d_2(\mathbf{x}, \mathbf{y})$.

Example

Let $\mathbf{x} = (2, -1, 3)^{\mathsf{T}}$ and $\mathbf{y} = (-1, 1, 3)^{\mathsf{T}}$. Compute $d_1(\mathbf{x}, \mathbf{y})$ and $d_2(\mathbf{x}, \mathbf{y})$.

Answer: We have

$$d_1(\mathbf{x}, \mathbf{y}) = |2 - (-1)| + |-1 - 1| + |3 - 3| = 3 + 2 + 0 = 5$$

$$d_2(\mathbf{x}, \mathbf{y}) = \sqrt{(2 - (-1))^2 + (-1 - 1)^2 + (3 - 3)^2} = \sqrt{9 + 4 + 0} = \sqrt{13}$$

Vector normalization

Note that general vectors may have different magnitudes. For a more consistent analysis, it is common to normalize the vectors to unit length:

$$\hat{\mathbf{x}} = \frac{\mathbf{x}}{\|\mathbf{x}\|_2}$$

It is easy to see that we have

$$\|\hat{\mathbf{x}}\|_2 = \left\| \frac{\mathbf{x}}{\|\mathbf{x}\|_2} \right\|_2 = \frac{\|\mathbf{x}\|_2}{\|\mathbf{x}\|_2} = 1$$

In other words, normalization preserves direction and removes magnitude.
Example: For $\mathbf{x} = (3, 4)^\mathsf{T}$, we have

$$\|\mathbf{x}\|_2 = \sqrt{3^2 + 4^2 = 5}$$

and so

$$\hat{\mathbf{x}} = \frac{\mathbf{x}}{\|\mathbf{x}\|_2} = \frac{(3, 4)^\mathsf{T}}{5} = \left(\frac{3}{5}, \frac{4}{5}\right)^\mathsf{T}$$

Vector normalization

Using normalized vectors can help us simplify many expressions.

For example, for nonzero $\mathbf{u}, \mathbf{v}$, we first get the normalized vectors

$$\hat{\mathbf{u}} = \frac{\mathbf{u}}{\|\mathbf{u}\|_2}, \quad \hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|_2}.$$

Then the cosine similarity is given by

$$\cos\theta = \frac{\hat{\mathbf{u}} \cdot \hat{\mathbf{v}}}{\|\hat{\mathbf{u}}\| \, \|\hat{\mathbf{v}}\|} = \frac{\hat{\mathbf{u}} \cdot \hat{\mathbf{v}}}{(1)(1)} = \hat{\mathbf{u}} \cdot \hat{\mathbf{v}}.$$

In other words, the cosine similarity becomes a simple dot product after normalization.

Vector normalization

Another simplification achieved by vector normalization is as follows.

If $\|\mathbf{u}\|_2 = \|\mathbf{v}\|_2 = 1$, then

$$\|\mathbf{u} - \mathbf{v}\|_2^2 = \|\mathbf{u}\|_2^2 + \|\mathbf{v}\|_2^2 - 2(\mathbf{u} \cdot \mathbf{v}) = 2 - 2(\mathbf{u} \cdot \mathbf{v}) = 2(1 - \cos\theta).$$

After rearranging the terms, we can get

$$\cos\theta = 1 - \frac{1}{2}\|\mathbf{u} - \mathbf{v}\|_2^2.$$

Therefore, maximizing cosine similarity is equivalent to minimizing Euclidean distance among unit vectors.

Remark: Without unit normalization, the extra length terms will alter the value and hence the above equivalence may not hold.

Example

Given $\mathbf{q} = (1, 0)^\mathsf{T}$, $\mathbf{a} = (10, 0)^\mathsf{T}$ and $\mathbf{b} = (1, 1)^\mathsf{T}$.

- Which one among $\mathbf{a}, \mathbf{b}$ is closer to $\mathbf{q}$ under Euclidean distance?
- Which one among $\mathbf{a}, \mathbf{b}$ has larger cosine similarity with $\mathbf{q}$?
- Is there any disagreement?

Example

Given $\mathbf{q} = (1, 0)^T$, $\mathbf{a} = (10, 0)^T$ and $\mathbf{b} = (1, 1)^T$.

- Which one among $\mathbf{a}, \mathbf{b}$ is closer to $\mathbf{q}$ under Euclidean distance?
- Which one among $\mathbf{a}, \mathbf{b}$ has larger cosine similarity with $\mathbf{q}$?
- Is there any disagreement?

Answer: Note that

$$\|\mathbf{q} - \mathbf{a}\|_2 = \sqrt{(1-10)^2 + (0-0)^2} = 9, \qquad \|\mathbf{q} - \mathbf{b}\|_2 = \sqrt{(1-1)^2 + (0-1)^2} = 1.$$

Therefore, b is close to $\mathbf{q}$ under Euclidean distance.

As for the cosine similarity,

$$\text{Cosine similarity between } \mathbf{q} \text{ and } \mathbf{a} \ = \frac{\mathbf{q} \cdot \mathbf{a}}{\|\mathbf{q}\|_2 \|\mathbf{a}\|_2} = \frac{10}{(1)(10)} = 1,$$

$$\text{Cosine similarity between } \mathbf{q} \text{ and } \mathbf{b} \ = \frac{\mathbf{q} \cdot \mathbf{b}}{\|\mathbf{q}\|_2 \|\mathbf{b}\|_2} = \frac{1}{(1)(\sqrt{2})} = \frac{1}{\sqrt{2}}$$

Therefore, $\mathbf{a}$ has larger cosine similarity with $\mathbf{q}$.

Example

Given $\mathbf{q} = (1, 0)^\mathsf{T}$, $\mathbf{a} = (10, 0)^\mathsf{T}$ and $\mathbf{b} = (1, 1)^\mathsf{T}$.

- Which one among $\mathbf{a}, \mathbf{b}$ is closer to $\mathbf{q}$ under Euclidean distance?
- Which one among $\mathbf{a}, \mathbf{b}$ has larger cosine similarity with $\mathbf{q}$?
- Is there any disagreement?

Answer: (continued)

Note that there is disagreement, as the Euclidean distance measures the absolute distance between two points, while cosine similarity measures the angle between two vectors and only consider the direction.

Dot product and orthogonality

We say that two vectors $\mathbf{u}, \mathbf{v}$ are **orthogonal** if they satisfy

$$\boxed{\mathbf{u} \cdot \mathbf{v} = 0}$$

For nonzero vectors, this means a right angle in Euclidean geometry.

Example: For $\mathbf{u} = (1, 1)^T$ and $\mathbf{v} = (1, -1)^T$, we have

$$\mathbf{u} \cdot \mathbf{v} = (1)(1) + (1)(-1) = 1 - 1 = 0.$$

Remark:

- The zero vector $\mathbf{0}$ is orthogonal to all vectors, as $\mathbf{0} \cdot \mathbf{v} = 0$.
- A non-zero vector will NOT be orthogonal to itself, as $\mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2 > 0$ for non-zero vector $\mathbf{v}$.

Properties of orthogonal vectors

If $\mathbf{u}$ and $\mathbf{v}$ are orthogonal vectors, then

$$\|\mathbf{u}+\mathbf{v}\|_2^2 = \|\mathbf{u}\|_2^2 + \|\mathbf{v}\|_2^2 .$$

Reason: We have

$$\|\mathbf{u}+\mathbf{v}\|_2^2 = (\mathbf{u}+\mathbf{v})\cdot(\mathbf{u}+\mathbf{v}) = \mathbf{u}\cdot\mathbf{u}+\mathbf{u}\cdot\mathbf{v}+\mathbf{v}\cdot\mathbf{u}+\mathbf{v}\cdot\mathbf{v} = \|\mathbf{u}\|_2^2 + 2(\mathbf{u}\cdot\mathbf{v}) + \|\mathbf{v}\|_2^2$$

Since $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, we have $\mathbf{u}\cdot\mathbf{v}=0$ and hence $\|\mathbf{u}+\mathbf{v}\|_2^2 = \|\mathbf{u}\|_2^2 + \|\mathbf{v}\|_2^2$.

More generally, if $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are mutually orthogonal (i.e., every pair of them is orthogonal), then we have

$$\|\mathbf{v}_1+\mathbf{v}_2+\cdots+\mathbf{v}_n\|_2^2 = \|\mathbf{v}_1\|_2^2 + \cdots + \|\mathbf{v}_n\|_2^2 .$$

Orthonormal vectors

More generally, we say that the vectors $\mathbf{q}_1, \ldots, \mathbf{q}_k$ are **orthonormal** if for all i, j, we have

$$\mathbf{q}_i \cdot \mathbf{q}_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

In other words,

- All the vectors have unit length
- They are pairwise orthogonal

If we put them into a matrix $Q = [\mathbf{q}_1 \ \cdots \ \mathbf{q}_k]$, we get

$$Q^\mathsf{T} Q = I \quad \text{(the identity matrix)}$$

Such a matrix satisfying the above property is called an orthogonal matrix.

Example

Show that the vectors

$$\mathbf{q}_1 = \begin{pmatrix} 1/\sqrt{2} \\ 1/2 \\ 1/2 \end{pmatrix}, \quad \mathbf{q}_2 = \begin{pmatrix} -1/\sqrt{2} \\ 1/2 \\ 1/2 \end{pmatrix}, \quad \mathbf{q}_3 = \begin{pmatrix} 0 \\ -1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

are orthonormal vectors.

Example

Show that the vectors

$$\mathbf{q}_1 = \begin{pmatrix} 1/\sqrt{2} \\ 1/2 \\ 1/2 \end{pmatrix}, \quad \mathbf{q}_2 = \begin{pmatrix} -1/\sqrt{2} \\ 1/2 \\ 1/2 \end{pmatrix}, \quad \mathbf{q}_3 = \begin{pmatrix} 0 \\ -1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

are orthonormal vectors.

Answer: We can check that

$$\|\mathbf{q}_1\|_2 = \sqrt{(1/\sqrt{2})^2 + (1/2)^2 + (1/2)^2} = \sqrt{1/2 + 1/4 + 1/4} = 1,$$

$$\|\mathbf{q}_2\|_2 = \sqrt{(-1/\sqrt{2})^2 + (1/2)^2 + (1/2)^2} = \sqrt{1/2 + 1/4 + 1/4} = 1,$$

$$\|\mathbf{q}_3\|_2 = \sqrt{(0)^2 + (-1/\sqrt{2})^2 + (1/\sqrt{2})^2} = \sqrt{0 + 1/2 + 1/2} = 1,$$

so all of them have unit length.

Example

Show that the vectors

$$\mathbf{q}_1 = \begin{pmatrix} 1/\sqrt{2} \\ 1/2 \\ 1/2 \end{pmatrix}, \quad \mathbf{q}_2 = \begin{pmatrix} -1/\sqrt{2} \\ 1/2 \\ 1/2 \end{pmatrix}, \quad \mathbf{q}_3 = \begin{pmatrix} 0 \\ -1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

are orthonormal vectors.

Answer: (continued) Also,

$$\mathbf{q}_1 \cdot \mathbf{q}_2 = (1/\sqrt{2})(-1/\sqrt{2}) + (1/2)(1/2) + (1/2)(1/2) = -1/2 + 1/4 + 1/4 = 0,$$

$$\mathbf{q}_1 \cdot \mathbf{q}_3 = (1/\sqrt{2})(0) + (1/2)(-1/\sqrt{2}) + (1/2)(1/\sqrt{2}) = 0 - 1/(2\sqrt{2}) + 1/(2\sqrt{2}) = 0,$$

$$\mathbf{q}_2 \cdot \mathbf{q}_3 = (-1/\sqrt{2})(0) + (1/2)(-1/\sqrt{2}) + (1/2)(1/\sqrt{2}) = 0 - 1/(2\sqrt{2}) + 1/(2\sqrt{2}) = 0,$$

so they are all orthogonal.

We conclude that they are orthonormal vectors.

Example

Show that the vectors

$$\mathbf{q}_1 = \begin{pmatrix} 1/\sqrt{2} \\ 1/2 \\ 1/2 \end{pmatrix}, \quad \mathbf{q}_2 = \begin{pmatrix} -1/\sqrt{2} \\ 1/2 \\ 1/2 \end{pmatrix}, \quad \mathbf{q}_3 = \begin{pmatrix} 0 \\ -1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

are orthonormal vectors.

Remark: One can also easily check that the matrix

$$Q = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/2 & 1/2 & -1/\sqrt{2} \\ 1/2 & 1/2 & 1/\sqrt{2} \end{pmatrix}$$

is an orthogonal matrix.

Example

Another typical example of orthogonal matrix is the rotation matrix $\begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}$ which can be used to rotate a vector in $\mathbb{R}^2$. We can easily check that the vectors

$$\begin{pmatrix} \cos t \\ -\sin t \end{pmatrix}, \quad \begin{pmatrix} \sin t \\ \cos t \end{pmatrix}$$

are orthonormal vectors. Note that it is also commonly used in AI for data augmentation (e.g., rotating training images by specific or random angles to help machine learning models recognize objects from any orientation).

Orthogonal projection

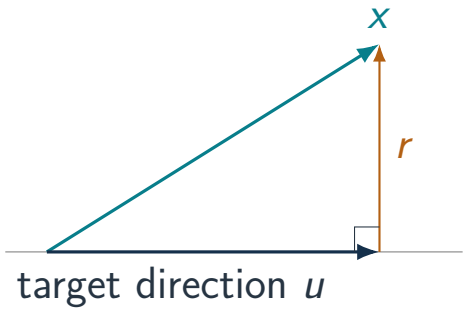
Note that orthogonal vectors are useful in AI as they represent completely independent information (with cosine similarity $= 0$).
They are commonly used in:

- Dimensionality reduction
- Data compression
- Neural network training
- ...

However, we do not always have vectors that are orthogonal by default. Therefore, it is important to have a method to transform general vectors into orthogonal vectors.
To achieve this, we can utilize the concept of orthogonal projection.

Orthogonal projection

Suppose we have a vector $\mathbf{x}$ and a target direction $\mathbf{u}$. The goal of orthogonal projection is to project $\mathbf{x}$ along the direction of $\mathbf{u}$ and discard other components.



More mathematically, the goal is to decompose $\mathbf{x}$ into

$$\mathbf{x} = \alpha\mathbf{u} + \mathbf{r},$$

where $\mathbf{r}$ is orthogonal to $\mathbf{u}$ and α is a scalar to be determined.

Orthogonal projection

Since $\mathbf{r}$ is desired to be orthogonal to $\mathbf{u}$, we have

$$0 = \mathbf{r} \cdot \mathbf{u} = (\mathbf{x} - \alpha\mathbf{u}) \cdot \mathbf{u} = \mathbf{x} \cdot \mathbf{u} - \alpha(\mathbf{u} \cdot \mathbf{u})$$

and hence

$$\alpha = \frac{\mathbf{x} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}.$$

Therefore, the orthogonal projection of $\mathbf{x}$ along $\mathbf{u}$ is given by

$$\boxed{\text{proj}_{\mathbf{u}}(\mathbf{x}) = \frac{\mathbf{u} \cdot \mathbf{x}}{\mathbf{u} \cdot \mathbf{u}}\mathbf{u}}$$

In particular, if u is a unit vector, we can further simplify the above as

$$\text{proj}_{\mathbf{u}}(\mathbf{x}) = (\mathbf{u} \cdot \mathbf{x})\mathbf{u}$$

Example

For $\mathbf{x} = (3, 1)^T$ and $\mathbf{u} = (1, 1)^T$, we have

$$\alpha = \frac{3(1) + 1(1)}{(1)(1) + (1)(1)} = 2$$

and

$$\mathbf{p} = \text{proj}_{\mathbf{u}}(\mathbf{x}) = 2\mathbf{u} = (2, 2)^T$$

Also,

$$\mathbf{r} = \mathbf{x} - \mathbf{p} = (3, 1)^T - (2, 2)^T = (1, -1)^T$$

We can easily check that

$$\mathbf{r} \cdot \mathbf{u} = (1)(1) + (-1)(1) = 0$$

Also,

$$\|\mathbf{x}\|_2^2 = 10 = 8 + 2 = \|\mathbf{p}\|_2^2 + \|\mathbf{r}\|_2^2$$

Projection is the nearest point on the line

We can show that the projection is the nearest point on the target line.

For any other point $c\mathbf{u}$ on the line, we have

$$\mathbf{x} - c\mathbf{u} = (\mathbf{x} - \mathbf{p}) + (\mathbf{p} - c\mathbf{u}) = \mathbf{r} + (\mathbf{p} - c\mathbf{u}).$$

The two terms are orthogonal. Hence

$$\|\mathbf{x} - c\mathbf{u}\|_2^2 = \|\mathbf{r}\|_2^2 + \|\mathbf{p} - c\mathbf{u}\|_2^2 \geq \|\mathbf{r}\|_2^2 .$$

Equality holds at $c\mathbf{u} = \mathbf{p}$.

In other words, projection **minimizes squared distance** to the line.

Example

Let $\mathbf{x} = (2, 3)^\top$ and $\mathbf{u} = (1, 0)^\top$. Suppose we want to project $\mathbf{x}$ along the direction of $\mathbf{u}$.

- Compute the projection $\mathbf{p}$ and residual $\mathbf{r}$.
- Verify $\mathbf{u} \cdot \mathbf{r} = 0$.
- Compare the distance from $\mathbf{x}$ to $\mathbf{p}$ with the distance to the other line point $(5, 0)^\top$.

Example

Let $\mathbf{x} = (2, 3)^\mathsf{T}$ and $\mathbf{u} = (1, 0)^\mathsf{T}$. Suppose we want to project $\mathbf{x}$ along the direction of $\mathbf{u}$.

- Compute the projection $\mathbf{p}$ and residual $\mathbf{r}$.
- Verify $\mathbf{u} \cdot \mathbf{r} = 0$.
- Compare the distance from $\mathbf{x}$ to $\mathbf{p}$ with the distance to the other line point $(5, 0)^\mathsf{T}$.

Answer: We have

$$\mathbf{p} = (2, 0)^\mathsf{T}, \quad \mathbf{r} = (0, 3)^\mathsf{T}$$

and

$$\mathbf{u} \cdot \mathbf{r} = (1)(0) + (0)(3) = 0.$$

Also,

$$\|\mathbf{x} - \mathbf{p}\|_2 = 3, \qquad \left\|\mathbf{x} - (5, 0)^\mathsf{T}\right\|_2 = \sqrt{(-3)^2 + 3^2} = \sqrt{18} > 3.$$

We can see that moving away from the projection adds a horizontal error while leaving the vertical error unchanged.

Projection as a matrix operation

Note that projection of $\mathbf{x}$ along the direction of $\mathbf{u}$ can be regarded as a matrix operation:

$$p = P\mathbf{x}, \text{ where } P = \frac{\mathbf{u}\mathbf{u}^\mathsf{T}}{\mathbf{u}^\mathsf{T}\mathbf{u}}$$

Example: For $\mathbf{u} = (1, 1)^\mathsf{T}$,

$$P = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

Every output has equal coordinates and therefore lies on the line through $\mathbf{u}$.

Projection as a matrix operation

Also, projecting twice changes nothing further.

Since $P\mathbf{x}$ is already on the target line, we have

$$P(P\mathbf{x}) = P\mathbf{x}, \qquad P^2 = P.$$

Another property is that the projection matrix P is symmetric:

$$P^{\mathsf{T}} = \left(\frac{\mathbf{u}\mathbf{u}^{\mathsf{T}}}{\mathbf{u}^{\mathsf{T}}\mathbf{u}}\right)^{\mathsf{T}} = \frac{1}{\mathbf{u}^{\mathsf{T}}\mathbf{u}}(\mathbf{u}\mathbf{u}^{\mathsf{T}})^{\mathsf{T}} = \frac{1}{\mathbf{u}^{\mathsf{T}}\mathbf{u}}(\mathbf{u}^{\mathsf{T}})^{\mathsf{T}}\mathbf{u}^{\mathsf{T}} = \frac{1}{\mathbf{u}^{\mathsf{T}}\mathbf{u}}\mathbf{u}\mathbf{u}^{\mathsf{T}} = P.$$

Here, we use the properties that $(AB)^{\mathsf{T}} = B^{\mathsf{T}}A^{\mathsf{T}}$ and $(A^{\mathsf{T}})^{\mathsf{T}} = A$.

Gram–Schmidt process and QR factorization

Goal: Given n linearly independent real column vectors $\{\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n\}$, construct an **orthonormal** set of vectors $\{\mathbf{q}_1, \mathbf{q}_2, \ldots, \mathbf{q}_n\}$.

Key idea:

- Start with a set of non-orthogonal vectors
- Systematically remove the overlapping parts (the projections) one by one, leaving behind a clean, mutually perpendicular set of vectors that span the exact same space

For instance, we can take $\mathbf{v}_1 = \mathbf{a}_1$ as the first vector

Then, to modify $\mathbf{a}_2$ so that the modified vector is orthogonal to $\mathbf{v}_1$, we can construct

$$\mathbf{v}_2 = \mathbf{a}_2 - \frac{\mathbf{v}_1 \cdot \mathbf{a}_2}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1$$

Note that:

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = \mathbf{v}_1 \cdot \left(\mathbf{a}_2 - \frac{\mathbf{v}_1 \cdot \mathbf{a}_2}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1\right) = \mathbf{v}_1 \cdot \mathbf{a}_2 - \frac{\mathbf{v}_1 \cdot \mathbf{a}_2}{\mathbf{v}_1 \cdot \mathbf{v}_1}(\mathbf{v}_1 \cdot \mathbf{v}_1) = \mathbf{v}_1 \cdot \mathbf{a}_2 - \mathbf{v}_1 \cdot \mathbf{a}_2 = 0$$

So the newly constructed $\mathbf{v}_2$ is orthogonal to $\mathbf{v}_1$

Gram–Schmidt process and QR factorization

Analogously, we can construct the third vector

$$\mathbf{v}_3 = \mathbf{a}_3 - \frac{\mathbf{v}_1 \cdot \mathbf{a}_3}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{v}_2 \cdot \mathbf{a}_3}{\mathbf{v}_2 \cdot \mathbf{v}_2}\mathbf{v}_2$$

In other words, we subtract $\mathbf{a}_3$ by the projection of it onto $\mathbf{v}_1$ and the projection of it onto $\mathbf{v}_2$. We can then further check that

$$\mathbf{v}_1 \cdot \mathbf{v}_3 = \mathbf{v}_1 \cdot \mathbf{a}_3 - \frac{\mathbf{v}_1 \cdot \mathbf{a}_3}{\mathbf{v}_1 \cdot \mathbf{v}_1}(\mathbf{v}_1 \cdot \mathbf{v}_1) - \frac{\mathbf{v}_2 \cdot \mathbf{a}_3}{\mathbf{v}_2 \cdot \mathbf{v}_2}(\mathbf{v}_1 \cdot \mathbf{v}_2) = \mathbf{v}_1 \cdot \mathbf{a}_3 - \mathbf{v}_1 \cdot \mathbf{a}_3 - 0 = 0$$

and similarly

$$\mathbf{v}_2 \cdot \mathbf{v}_3 = 0$$

Therefore, $\mathbf{v}_3$ is orthogonal to $\mathbf{v}_1$ and $\mathbf{v}_2$.

Gram–Schmidt process and QR factorization

More generally, we have the following procedure of the **Gram–Schmidt process**:

1. First get an **orthogonal** set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$:

$$\begin{aligned}
\mathbf{v}_1 &= \mathbf{a}_1, \\
\mathbf{v}_2 &= \mathbf{a}_2 - \frac{\mathbf{v}_1 \cdot \mathbf{a}_2}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 \\
&\vdots \\
\mathbf{v}_n &= \mathbf{a}_n - \frac{\mathbf{v}_1 \cdot \mathbf{a}_n}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{v}_2 \cdot \mathbf{a}_n}{\mathbf{v}_2 \cdot \mathbf{v}_2}\mathbf{v}_2 - \cdots - \frac{\mathbf{v}_{n-1} \cdot \mathbf{a}_n}{\mathbf{v}_{n-1} \cdot \mathbf{v}_{n-1}}\mathbf{v}_{n-1}.
\end{aligned}$$

2. Get an **orthonormal** set of vectors $\{\mathbf{q}_1, \mathbf{q}_2, \ldots, \mathbf{q}_n\}$:

$$\mathbf{q}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|_2}, \ \mathbf{q}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|_2}, \ \ldots, \ \mathbf{q}_n = \frac{\mathbf{v}_n}{\|\mathbf{v}_n\|_2}.$$

Gram–Schmidt process and QR factorization

If $A = \left(\begin{array}{c|c|c|c} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{array} \right)$ is an $m \times n$ real matrix where $m \geq n$ and its columns $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ are linearly independent, then the above process gives the **QR factorization** of A:

$$A = QR,$$

where $Q = \left(\begin{array}{c|c|c|c} \mathbf{q}_1 & \mathbf{q}_2 & \cdots & \mathbf{q}_n \end{array} \right)$ is an $m \times n$ matrix with orthonormal columns (i.e. with $Q^T Q = I$) and $R = \begin{pmatrix} r_{11} & r_{12} & \ldots & r_{1n} \\ 0 & r_{22} & \ldots & r_{2n} \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \ldots & r_{nn} \end{pmatrix}$ is an $n \times n$ invertible upper triangular matrix, with $r_{ij} = \mathbf{q}_i \cdot \mathbf{a}_j$ for all $i \leq j$.

Example

Find the QR factorization of the matrix $A = \begin{pmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 2 & 0 & 4 \end{pmatrix}$.

Example

Find the QR factorization of the matrix $A = \begin{pmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 2 & 0 & 4 \end{pmatrix}$.

Answer: We first get three orthogonal vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$:

$$\mathbf{v}_1 = \mathbf{a}_1 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix},$$

$$\mathbf{v}_2 = \mathbf{a}_2 - \frac{\mathbf{v}_1 \cdot \mathbf{a}_2}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} - \frac{0}{6}\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix},$$

$$\mathbf{v}_3 = \mathbf{a}_3 - \frac{\mathbf{v}_1 \cdot \mathbf{a}_3}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{v}_2 \cdot \mathbf{a}_3}{\mathbf{v}_2 \cdot \mathbf{v}_2}\mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix} - \frac{9}{6}\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}.$$

Example

Find the QR factorization of the matrix $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 2 & 0 & 4 \end{pmatrix}$.

Answer: (continued) Then, we get three orthonormal vectors $\{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3\}$:

$$\mathbf{q}_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \ \mathbf{q}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \ \mathbf{q}_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}.$$

Finally, we have

$$A = QR = \begin{pmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \end{pmatrix} \begin{pmatrix} \sqrt{6} & 0 & \frac{3\sqrt{3}}{\sqrt{2}} \\ 0 & \sqrt{2} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \sqrt{3} \end{pmatrix}.$$

Overdetermined systems and normal equations

Consider an overdetermined system

$$\begin{bmatrix} & \\ & A \\ & \end{bmatrix} \begin{bmatrix} \\ \mathbf{x} \\ \\ \end{bmatrix} = \begin{bmatrix} \\ \mathbf{b} \\ \\ \end{bmatrix},$$

where A is an $m \times n$ matrix with $m > n$.

- In general, more constraints than variables $\Rightarrow$ no solution!
- We look for $\mathbf{x}$ that makes $\|A\mathbf{x} - \mathbf{b}\|$ as small as possible
- More explicitly, we look for the **least-squares solution** of $A\mathbf{x} = \mathbf{b}$, i.e. a vector $\mathbf{x}$ that satisfies

$$\|A\mathbf{x} - \mathbf{b}\|^2 \leq \|A\mathbf{y} - \mathbf{b}\|^2 \text{ for all } \mathbf{y} \in \mathbb{R}^n$$

To achieve this, we consider the **normal equation**

$$\boxed{A^T A\mathbf{x} = A^T \mathbf{b}}$$

Overdetermined systems and normal equations

Theorem:

The solution to the normal equation $A^T A\mathbf{x} = A^T\mathbf{b}$ provides the least-squares solution to $A\mathbf{x} = \mathbf{b}$.

Proof: We need to show that if $\mathbf{x}$ satisfies $A^T A\mathbf{x} = A^T\mathbf{b}$, then $\|A\mathbf{x} - \mathbf{b}\| \leq \|A\mathbf{y} - \mathbf{b}\|$ for any vector $\mathbf{y}$. Let $\mathbf{r_x} = A\mathbf{x} - \mathbf{b}$ and $\mathbf{r_y} = A\mathbf{y} - \mathbf{b}$. From the normal equation, we have

$$A^T\mathbf{r_x} = 0 \Rightarrow \mathbf{r_x}^T A = 0$$

Then,

$$\mathbf{r_y} = A\mathbf{y} - \mathbf{b} = (A\mathbf{x} - \mathbf{b}) + (A\mathbf{y} - A\mathbf{x}) = \mathbf{r_x} + A(\mathbf{y} - \mathbf{x}).$$

Therefore,

$$\begin{aligned}\|\mathbf{r_y}\|^2 &= \langle \mathbf{r_x} + A(\mathbf{y} - \mathbf{x}), \mathbf{r_x} + A(\mathbf{y} - \mathbf{x})\rangle \\ &= \|\mathbf{r_x}\|^2 + \mathbf{r_x}^T A(\mathbf{y} - \mathbf{x}) + (A(\mathbf{y} - \mathbf{x}))^T\mathbf{r_x} + \|A(\mathbf{y} - \mathbf{x})\|^2 \\ &= \|\mathbf{r_x}\|^2 + 0 + 0 + \|A(\mathbf{y} - \mathbf{x})\|^2 \geq \|\mathbf{r_x}\|^2.\end{aligned}$$

Overdetermined systems and QR factorization

Let A be an $m \times n$ matrix ($m \geq n$) and $\mathbf{b}$ be an $m \times 1$ vector.
If we have the QR factorization

$$A = QR,$$

then the least-squares solution to $A\mathbf{x} = \mathbf{b}$ is given by

$$\boxed{R\mathbf{x} = Q^T\mathbf{b}}.$$

We can then solve for $\mathbf{x}$ using backward substitution.

(Proof: We have $A^TA\mathbf{x} = A^T\mathbf{b} \Rightarrow R^TQ^TQR\mathbf{x} = R^TQ^T\mathbf{b}$. Since $Q^TQ = I$ and R^T is invertible, the result follows easily.)

Gram matrices

For rows $\mathbf{x}_i^\top$ of X, we define the Gram matrix as

$$G = XX^\top, \qquad G_{ij} = \mathbf{x}_i \cdot \mathbf{x}_j = \mathbf{x}_i^\top \mathbf{x}_j.$$

Note that:

- G is an $n \times n$ symmetric matrix.
- It collects pairwise dot products and hence can be used to encode the cosine similarity.

Remark: If the rows have unit length, its entries are pairwise cosine similarities. If not, then we may not directly get the cosine similarities and need some further procedures.

Example:

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \quad \Longrightarrow \quad XX^\top = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}.$$

Diagonal entries are squared row lengths. The third row is not a unit vector, so this raw matrix is not a cosine-similarity matrix.

Example

Suppose

$$G = \begin{bmatrix} 4 & 0 & 2 \\ 0 & 9 & 3 \\ 2 & 3 & 2 \end{bmatrix} = XX^{\mathsf{T}}.$$

- Find the lengths of the three row vectors of X.
- Which two rows are orthogonal?
- Find the cosine similarity between rows 1 and 3.

Example

Suppose

$$G = \begin{bmatrix} 4 & 0 & 2 \\ 0 & 9 & 3 \\ 2 & 3 & 2 \end{bmatrix} = XX^{\mathsf{T}}.$$

- Find the lengths of the three row vectors of X.
- Which two rows are orthogonal?
- Find the cosine similarity between rows 1 and 3.

Answer:
Lengths are 2, 3 and $\sqrt{2}$ because $G_{ii} = \|\mathbf{x}_i\|_2^2$.
Rows 1 and 2 are orthogonal since $G_{12} = 0$.
The cosine similarity is

$$\frac{G_{13}}{\sqrt{G_{11}}\sqrt{G_{33}}} = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}}.$$

Next time

More on linear algebra:

- Rank, inverse, span
- Eigen-decomposition
- SVD
- Conditioning of matrices
- Applications to graph data